Question			Answer	Marks	AO	Guidance
11	**(a)**		Nina's, because hers is the largest sample size **oe**	**B1**	**2.2a**	allow eg Nina's, because with a larger sample size the probabilities get closer to the true probabilities **oe**
				[1]		
11	**(b)**		$11p + kp = 1$ $p = \frac{1}{11+k}$	**M1** **A1**	**3.1a** **1.1**	
				[2]		
11	**(c)**		$their\ \frac{1}{11+k} \times k$ or $their\ \frac{1}{11+k} \times 120$ $120 \times their\ \frac{k}{11+k}$ $\frac{120k}{11+k}$ **oe**	**M1** **M1** **A1**	**2.1** **1.2** **1.1**	multiply by k or by 120; may be embedded multiplying by both k and 120
				[3]		
11	**(d)**		32 = their $\frac{120k}{11+k}$ **oe** $k = 4$	**M1** **A1**	**1.1** **1.1**	
				[2]		

Question			Answer	Marks	AO	Guidance
			Alternatively $11p = 1 - \frac{32}{120}$ may be implied by $p = \frac{1}{15}$ (from $\left(P(X \neq 12)\right)$) $k = 4$	**M1** **A1**		**or** $\frac{kf}{120} = \frac{32}{120}$ (from $11f = 120 - 32 = 88$ so $f = 8$ and so $kp = \cdots$) $k = 4$
11	**(e)**		$Y \sim \text{B}\left(30, their\ \frac{4}{11+4}\right)$ or $Y \sim \text{B}\left(30, \frac{32}{120}\right)$ used to find P($Y = 8$) 0.16 – 0.163 **BC**	**M1** **A1**	**3.1a** **1.1**	Y is the number of 12s obtained in 30 rolls; allow **B2** for 0.1628 – 0.163 unsupported
				[2]		